Math 237B Final

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March 15, 2026

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Problem 1. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes such that $g \circ f$ is proper, g is quasi-projective and f is surjective. Prove or disprove that g is proper.

Solution. We'll prove that g is proper. We'll prove relevant statements in the following order:
(a) g is separated, and f is proper.
(b) g is of finite type.
(c) g is universally closed.
Finally, use these results we can conclude that g is proper.

Proof of (a):

Since $g : Y \to Z$ is quasi-projective, there exists some scheme Y', together with open immersion $\iota : Y \hookrightarrow Y'$ and projective morphism $\pi : Y' \to Z$, such that the following commutative diagram holds:

$$\begin{tikzcd} Y \arrow[r, hook, "\iota"] \arrow[d, "g"'] & Y' \arrow[dl, "\pi"] \\ Z & \end{tikzcd}$$

Which, recall that any open immersion is separated, and projective morphisms are proper (in particular separated), so $g = \pi \circ \iota$ is also separated.

As a consequence, since $g \circ f$ is proper, while g is separated, we concluded that f is proper also.

Proof of (b):

Let's show g is of finite type: Using the diagram in part (a), since $\pi : Y' \to Z$ is projective, it is proper (hence of finite type), so for any open affine subset $U \subseteq Z$ (say $U \cong \operatorname{Spec}(A)$), such that $\pi^{-1}(U) = \bigcup_{i=1}^n V_i$, where each V_i is affine open subset, and $V_i \cong \operatorname{Spec}(B_i)$ for some B_i that's finitely generated A-algebra.

Then, with open immersion $\iota : Y \hookrightarrow Y'$, one has $\iota^{-1}(\pi^{-1}(V)) = Y \cap f^{-1}(V) = \bigcup_{i=1}^n Y \cap V_i$. Since Y is open in Y', each $Y \cap V_i$ is an open subset of V_i under its subspace topology, hence each $Y \cap V_i = \bigcup_{j \in J_i} D(f_{ij})$ for some $f_{ij} \in B_i$ (running over some index set J_i). Because each $D(f_{ij}) \cong \operatorname{Spec}\left((B_i)_{f_{ij}}\right)$ (which $(B_i)_{f_{ij}}$ is a finitely generated B_i-algebra), then with B_i being a finitely generated A-algebra, $(B_i)_{f_{ij}}$ is a finitely generated A-algebra also.

As a result, one has $g^{-1}(U) = \iota^{-1}(\pi^{-1}(U)) = \bigcup_{i=1}^n Y \cap B_i = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$, where each $D(f_{ij}) \cong \operatorname{Spec}\left((B_i)_{f_{ij}}\right)$ has $(B_i)_{f_{ij}}$ being a finitely generated A-algebra.

Finally, since $g \circ f$ is finite type, it's quasi-compact, hence with affine open subset $U \subseteq Z$, one has $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$ being quasi-compact. However, since f is surjective, one has $g^{-1}(U) =$

$f(f^{-1}(g^{-1}(U)))$. Because it's an image of a quasi-compact set under continuous map, $g^{-1}(U)$ is quasi-compact. Hence, the previous cover for $g^{-1}(U)$, say $g^{-1}(U) = \bigcup_{i=1}^{n} \bigcup_{j \in J_i} D(f_{ij})$, can be chosen as finite. This shows that g is of finite type.

Proof of (c):

To prove g is universally closed, for any morphism of schemes $h : Z' \to Z$ consider the following commutative diagram of multiple fibre products:

$$\begin{tikzcd}
(Z' \times_Z Y) \times_Y X \arrow[r, "h''"] \arrow[d, two heads, "f'"'] & X \arrow[d, two heads, "f"] \\
Z' \times_Z Y \arrow[r, "h'"] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\
Z' \arrow[r, "h"'] & Z
\end{tikzcd}$$

Which:

- $g' : Z' \times_Z Y \to Z'$ denotes the base change of $g : Y \to Z$ by $h : Z' \to Z$.
- $f' : (Z' \times_Z Y) \times_Y X \twoheadrightarrow Z' \times_Z Y$ denotes the base change of $f : X \twoheadrightarrow Y$ by $h' : Z' \times_Z Y \to Y$ (the pullback of $h : Z' \to Z$), which the surjectivity of f ensures the surjectivity of f'. And, since part (a) shows f is proper (which is universally closed), then f' is closed.

Now, we ought to show $(Z' \times_Z Y) \times_Y X$ together with $g' \circ f'$ and h'' satisfies the universal property of the fibre product $Z' \times_Z X$. For any scheme W together with two morphisms $p_1 : W \to X$ and $p_2 : W \to Z'$, such that $(g \circ f) \circ p_1 = h \circ p_2$. Then, the universal property of $Z' \times_Z Y$ guarantees a unique morphism of schemes $\varphi : W \to Z' \times_Z Y$, such that $p_2 = g' \circ \varphi$, and $f \circ p_1 = h' \circ \varphi$, as follow:

$$\begin{tikzcd}
W \arrow[drr, bend left, "p_1"] \arrow[ddr, dotted, "\exists!\varphi"] \arrow[dddr, bend right, "p_2"'] & & \\
& (Z' \times_Z Y) \times_Y X \arrow[r, "h''"] \arrow[d, two heads, "f'"'] & X \arrow[d, two heads, "f"] \\
& Z' \times_Z Y \arrow[r, "h'"] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\
& Z' \arrow[r, "h"'] & Z
\end{tikzcd}$$

Furthermore, with the universal property of $(Z' \times_Z Y) \times_Y X$, the equality $f \circ p_1 = h' \circ \varphi$ guarantees a unique morphism of schemes $\psi : W \to (Z' \times_Z Y) \times_Y X$, such that $\varphi = f' \circ \psi$ and $p_1 = h'' \circ \psi$, as follow:

$$\begin{tikzcd}
W \arrow[dr, dotted, "\exists!\psi"] \arrow[ddr, dotted, "\exists!\varphi"'] \arrow[drr, bend left, "p_1"] \arrow[dddr, bend right, "p_2"'] & & \\
& (Z' \times_Z Y) \times_Y X \arrow[r, "h''"] \arrow[d, two heads, "f'"'] & X \arrow[d, two heads, "f"] \\
& Z' \times_Z Y \arrow[r, "h'"] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\
& Z' \arrow[r, "h"'] & Z
\end{tikzcd}$$

This proves both the existence and the uniqueness of the morphism $\psi : W \to (Z' \times_Z Y) \times_Y X$, such that $p_2 = g' \circ \varphi = (g' \circ f') \circ \psi$, and $p_1 = h'' \circ \psi$ (as any $\psi' : W \to (Z' \times_Z Y) \times_Y X$ with such property is enforced to be equal to ψ, based on individual fibre product's property). So, $(Z' \times_Z Y) \times_Y X$ indeed satisfies the universal property of $Z' \times_Z X$, showing that $g' \circ f'$ is the base change of the morphism $(g \circ f) : X \to Z$. Which, since $g \circ f$ is proper, $g' \circ f'$ is closed.

Finally, with f' and $g' \circ f'$ being closed, and f' being surjective, we claim that g' is closed:

For any closed subset $V \subseteq Z' \times_Z Y$, by the continuity of f', the subset $(f')^{-1}(V) \subseteq (Z' \times_Z Y) \times_Y X$ is closed. Hence, $(g' \circ f')\big((f')^{-1}(V)\big) \subseteq Z'$ is closed (by closeness of $g' \circ f'$).

Also, since f' is surjective, one simply has $f'\big((f')^{-1}(V)\big) = V$. Hence, the above set equality shows that $(g' \circ f')\big((f')^{-1}(V)\big) = g'(V) \subseteq Z'$ is closed. This proves that g' is closed.

With g' being based change of g over arbitrary morphism of schemes $h : Z' \to Z$, this shows that g is universally closed.

With all information above, we can conclude that g is separated, finite type, and universally closed, hence proper.

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Problem 2. Let $f : X \to S$ and $g : Y \to S$ be projective morphisms. Prove or disprove that the projection map $X \times_S Y \to S$ is projective.

Solution. We'll show that $X \times_S \to S$ is projective. We'll prove the following two statements:

(a) Base change of projective morphisms are projective.

(b) Composition of projective morphisms are projective.

Then, the two will help conclude the desired result.

Proof of (a):

First, we'll show that base change preserves projective spaces: More precisely, given any $n \in \mathbb{N}$, with morphisms $g : Y \to S$, the projective n-space over S, $\pi : \mathbb{P}^n_S \to S$ has fibre product $Y \times_S \mathbb{P}^n_S$ satisfies the universal property of $\mathbb{P}^n_Y$.

Indeed, with the definition of $\mathbb{P}^n_S := S \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$, we form the following commutative diagram:

$$\begin{tikzcd}
Y \times_S \mathbb{P}^n_S \arrow[r, "g'"] \arrow[d, "\pi'"'] & \mathbb{P}^n_S \arrow[r, "s'"] \arrow[d, "\pi"] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d, "\varphi"] \\
Y \arrow[r, "g"'] & S \arrow[r, "s"'] & \mathrm{Spec}(\mathbb{Z})
\end{tikzcd}$$

Where, both small squares are fibre product diagrams. Also, the morphism $s \circ g : Y \to \mathrm{Spec}(\mathbb{Z})$ is the unique morphism based on the terminal property of $\mathrm{Spec}(\mathbb{Z})$ in the category of schemes.

Now, given any scheme W together with two morphisms $p_1 : W \to \mathbb{P}^n_{\mathbb{Z}}$ and $p_2 : W \to Y$, such that $\varphi \circ p_1 = (s \circ g) \circ p_2$, then the universal property of $\mathbb{P}^n_S$ generates a unique morphism $\psi : W \to \mathbb{P}^n_S$, such that $p_1 = s' \circ \psi$, and $g \circ p_2 = \pi \circ \psi$, as follow:

$$\begin{tikzcd}
W \arrow[drrr, bend left, "p_1"] \arrow[drr, dotted, "\exists!\psi"] \arrow[ddr, bend right, "p_2"'] & & & \\
& Y \times_S \mathbb{P}^n_S \arrow[r, "g'"] \arrow[d, "\pi'"'] & \mathbb{P}^n_S \arrow[r, "s'"] \arrow[d, "\pi"] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d, "\varphi"] \\
& Y \arrow[r, "g"'] & S \arrow[r, "s"'] & \mathrm{Spec}(\mathbb{Z})
\end{tikzcd}$$

Then, with the universal property of $Y \times_S \mathbb{P}^n_S$, the second equality $g \circ p_2 = \pi \circ \psi$ guarantees a unique morphism $\rho : W \to Y \times_S \mathbb{P}^n_S$, such that $\psi = g' \circ \rho$, and $p_2 = \pi' \circ \rho$, as follow:

$$\begin{tikzcd}
W \arrow[drrr, bend left, "p_1"] \arrow[drr, dotted, "\exists!\psi"] \arrow[dr, dotted, "\exists!\rho"] \arrow[ddr, bend right, "p_2"'] & & & \\
& Y \times_S \mathbb{P}^n_S \arrow[r, "g'"] \arrow[d, "\pi'"'] & \mathbb{P}^n_S \arrow[r, "s'"] \arrow[d, "\pi"] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d, "\varphi"] \\
& Y \arrow[r, "g"'] & S \arrow[r, "s"'] & \mathrm{Spec}(\mathbb{Z})
\end{tikzcd}$$

Again, this proves both the existence and the uniqueness of the morphism $\rho : W \to Y \times_S \mathbb{P}^n_S$, such that $p_1 = s' \circ \psi = (s' \circ g') \circ \rho$, and $p_2 = \pi' \circ \rho$. Hence, $Y \times_S \mathbb{P}^n_S$ satisfies the universal property of $\mathbb{P}^n_Y := Y \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$, showing $Y \times_S \mathbb{P}^n_S \cong \mathbb{P}^n_Y$.

Now, we'll prove that base change preserves projective morphisms, again by the universality argument: Given a projective morphism $f : X \to S$, and another morphism $g : Y \to S$, we know there exists a suitable $n \in \mathbb{N}$, such that there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_S$ and the projection $\pi : \mathbb{P}^n_S \to S$, such that $f = \pi \circ \iota$.

Which, based on the previous argument, we'll simply denote $Y \times_S \mathbb{P}^n_S$ as $\mathbb{P}^n_Y$. Together with the previous diagrams, we have the following:

$$\begin{tikzcd}
X \times_{\mathbb{P}^n_S} \mathbb{P}^n_Y \arrow[r, "g''"] \arrow[d, hook, "\iota'"'] & X \arrow[d, hook, "\iota"] & \\
\mathbb{P}^n_Y \arrow[r, "g'"] \arrow[d, "\pi'"'] & \mathbb{P}^n_S \arrow[r, "s'"] \arrow[d, "\pi"] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d, "\varphi"'] \\
Y \arrow[r, "g"'] & S \arrow[r, "s"'] & \mathrm{Spec}(\mathbb{Z})
\end{tikzcd}$$

Which, we took the fibre product $X \times_{\mathbb{P}^n_S} \mathbb{P}^n_Y$, and the morphism $\iota' : X \times_{\mathbb{P}^n_S} \mathbb{P}^n_Y \hookrightarrow \mathbb{P}^n_Y$ is the pullback of the closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_S$, hence ι' is also a closed immersion.

Now, we claim that it satisfies the universal property of $X \times_S Y$: In general, the above proof of preserving projective spaces shows that successive fibre product preserves fibre product in the full generality. By modifying the labels / objects a bit, we indeed get a proof that $X \times_{\mathbb{P}^n_S} \mathbb{P}^n_Y$ satisfies the universal property of $X \times_S Y$ as fibre product, hence $X \times_{\mathbb{P}^n_S} \mathbb{P}^n_Y \cong X \times_S Y$.

Finally, by denoting $X \times_{\mathbb{P}^n_S} \mathbb{P}^n_Y$ as $X \times_S Y$, the above diagram reduces to the following fibre product diagram:

$$\begin{tikzcd}
X \times_S Y \arrow[r, "g''"] \arrow[d, hook, "\iota'"'] & X \arrow[d, "\iota"] \arrow[dd, bend left, "f"] \\
\mathbb{P}^n_Y \arrow[d, "\pi'"'] & \mathbb{P}^n_S \arrow[d, "\pi"] \\
Y \arrow[r, "g"'] & S
\end{tikzcd}$$

Which, the pullback of f is $\pi' \circ \iota' : X \times_S Y \to Y$, which is precisely a closed immersion into $\mathbb{P}^n_Y$, then a projection onto Y. This shows that the pullback of f is still projective, hence projective morphisms are preserved under base change.

Proof of (b):

Let $\rho : T \to U$ and $\varphi : U \to V$ be two projective morphisms, then there exists $n, m \in \mathbb{N}$, such that there exists open immersion $\iota_T : T \hookrightarrow \mathbb{P}^n_U$ and $\iota_U : U \hookrightarrow \mathbb{P}^m_V$, together with projections $\pi_{U,n} : \mathbb{P}^n_U \to U$ and $\pi_{V,m} : \mathbb{P}^m_V \to V$, such that the following diagram commutes:

$$\begin{tikzcd}
\mathbb{P}^n_U \arrow[dr, "\pi_{U,n}"] & \mathbb{P}^m_V \arrow[dr, "\pi_{V,m}"] & \\
T \arrow[u, hook, "\iota_T"] \arrow[r, "\rho"'] & U \arrow[u, hook, "\iota_U"'] \arrow[r, "\varphi"'] & V
\end{tikzcd}$$

Now, let's use some information from part (a): Recall that given any morphism $Y \to S$ and the projection $\mathbb{P}^n_S \to S$, the fibre product $Y \times_S \mathbb{P}^n_S \cong \mathbb{P}^n_Y$ (and it's true for arbitrary Y, S and $n \in \mathbb{N}$, as the proof is universal); then, from the above diagram, consider $\varphi : U \to V$ as the composition $U \xrightarrow{\iota_U} \mathbb{P}^m_V \xrightarrow{\pi_{V,m}} V$, and consider the projection $\pi_{V,n} : \mathbb{P}^n_V \to V$. Notice by the previous statementment, we can construct the projective space $\mathbb{P}^n_{\mathbb{P}^m_V} \cong \mathbb{P}^m_V \times_V \mathbb{P}^n_V$, as follow:

$$\begin{tikzcd}
 & \mathbb{P}^n_{\mathbb{P}^m_V} \arrow[r, "(\pi_{V,m})'"] \arrow[d, dash, "(\pi_{V,n})'"] & \mathbb{P}^n_V \arrow[d, "\pi_{V,n}"] \\
U \arrow[r, hook, "\iota_U"'] & \mathbb{P}^m_V \arrow[r, "\pi_{V,m}"'] & V
\end{tikzcd}$$

Which, the same argument shows that $\mathbb{P}^n_U \cong U \times_{\mathbb{P}^m_V} \mathbb{P}^n_{\mathbb{P}^m_V}$, as follow:

$$\begin{tikzcd}
\mathbb{P}^n_U \arrow[r, hook, "(\iota_U)'"] \arrow[d, "\pi_{U,n}"'] & \mathbb{P}^n_{\mathbb{P}^m_V} \arrow[r, "(\pi_{V,m})'"] \arrow[d, "(\pi_{V,n})'"] & \mathbb{P}^n_V \arrow[d, "\pi_{V,n}"] \\
U \arrow[r, hook, "\iota_U"'] & \mathbb{P}^m_V \arrow[r, "\pi_{V,m}"'] & V
\end{tikzcd}$$

Which, notice that $(\iota_U)' : \mathbb{P}^n_U \hookrightarrow \mathbb{P}^n_{\mathbb{P}^m_V}$ is a base change of the closed immersion $\iota_U : U \hookrightarrow \mathbb{P}^m_V$, hence is still a closed immersion; composing with the closed immersion $\iota_T : T \hookrightarrow \mathbb{P}^n_U$, we get a closed immesion $(\iota_U)' \circ \iota_T : T \hookrightarrow \mathbb{P}^n_{\mathbb{P}^m_V} \cong \mathbb{P}^m_V \times_V \mathbb{P}^n_V$, and with the projections $\pi_{V,n} \circ (\pi_{V,m})' : \mathbb{P}^n_{\mathbb{P}^m_V} \to V$, they compose to be the original morphism $T \xrightarrow{\rho} U \xrightarrow{\varphi} V$ (as $\left(\pi_{V,n} \circ (\pi_{V,m})'\right) \circ \left((\iota_U)' \circ \iota_T\right) = \left(\pi_{V,m} \circ \iota_U\right) \circ \left(\pi_{U,n} \circ \iota_T\right) = \varphi \circ \rho$ based on the diagrams).

Hence, to prove the composition $\varphi \circ \rho : T \to V$ is projective, it suffices to show that $\mathbb{P}^n_{\mathbb{P}^m_V} \cong \mathbb{P}^m_V \times_V \mathbb{P}^n_V$ embedds into some $\mathbb{P}^N_V$, and the projection backs to $\mathbb{V}$ recovers $\varphi \circ \rho$.

Let's first show that there exists a morphism $p : \mathbb{P}^m_V \times_V \mathbb{P}^n_V \to \mathbb{P}^m_{\mathbb{Z}} \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$. Consider the following diagram:

$$\begin{tikzcd}
\mathbb{P}^m_V \times_V \mathbb{P}^n_V \arrow[r, "(\pi_{V,m})'"] \arrow[d, "(\pi_{V,n})'"'] & \mathbb{P}^n_V \arrow[r] \arrow[d, "\pi_{V,n}"] & \mathbb{P}^n_{\mathbb{Z}} \arrow[dd] \\
\mathbb{P}^m_V \arrow[r, "\pi_{V,m}"'] \arrow[d] & V \arrow[dr] & \\
\mathbb{P}^m_{\mathbb{Z}} \arrow[rr] & & \mathrm{Spec}(\mathbb{Z})
\end{tikzcd}$$

Which, with each „square“ (two are deformed a bit) being fibre squares, the above diagram commutes. Hence, by the universal property of $\mathbb{P}^m_{\mathbb{Z}} \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$ guarantees a unique morphism $P : \mathbb{P}^m_V \times_V \mathbb{P}^n_V \to \mathbb{P}^m_{\mathbb{Z}} \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$.

Now, consider the Segre Embedding that is suggested in Hartshorne: Given $\mathbb{P}^m_{\mathbb{Z}} = \mathrm{Proj}(\mathbb{Z}[x_0, x_1, ..., x_m])$ and $\mathbb{P}^n_{\mathbb{Z}} = \mathrm{Proj}(\mathbb{Z}[y_0, y_1, ..., y_n])$, we cover the fibre product $\mathbb{P}^m_{\mathbb{Z}} \times_{\mathrm{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$ by the open affine subsets $D_+(x_i) \times_{\mathrm{Spec}(\mathbb{Z})} D_+(y_j)$ for $0 \le i \le m$ and $0 \le j \le n$. Which, with each $D_+(x_i) \cong \mathrm{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, \frac{x_1}{x_i}, ..., \frac{x_m}{x_i}\right]\right)$ and $D_+(y_j) \cong \mathrm{Spec}\left(\mathbb{Z}\left[\frac{y_0}{y_j}, \frac{y_1}{y_j}, ..., \frac{y_n}{y_j}\right]\right)$. Hence, their fibre product over $\mathrm{Spec}(\mathbb{Z})$ are given as follow:

$$\begin{aligned}
D_+(x_i) \times_{\mathrm{Spec}(\mathbb{Z})} D_+(y_j) &\cong \mathrm{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, ..., \frac{x_m}{x_i}\right] \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{y_0}{y_j}, ..., \frac{y_n}{y_j}\right]\right) \\
&\cong \mathrm{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, ..., \frac{x_m}{x_i}, \frac{y_0}{y_j}, ..., \frac{y_n}{y_j}\right]\right)
\end{aligned} \tag{2.1}$$

Then, with $N \coloneqq nm+n+m$, consider $\mathbb{P}^N_{\mathbb{Z}} = \operatorname{Proj}(\mathbb{Z}[z_0, ..., z_{ij}, ..., z_N])$. For any indeterminate z_{ij}, consider its fundamental open subset $D_+(z_{ij}) = \operatorname{Spec}\left(\mathbb{Z}\left[\frac{z_0}{z_{ij}}, ..., \frac{z_N}{z_{ij}}\right]\right)$, and a ring homomorphism $\psi_{ij} : \mathbb{Z}\left[\frac{z_0}{z_{ij}}, ..., \frac{z_N}{z_{ij}}\right] \to \mathbb{Z}\left[\frac{x_0}{x_i}, ..., \frac{x_m}{x_i}, \frac{y_0}{y_j}, ..., \frac{y_n}{y_j}\right]$ as follow:

$$\psi_{ij}\left(\frac{z_{i'j'}}{z_{ij}}\right) = \frac{x_{i'}}{x_i}\frac{y_{j'}}{y_j} \tag{2.2}$$

Notice that any $\frac{x_{i'}}{x_i}$ satisfies $\psi_{ij}\left(\frac{z_{i'j}}{z_{ij}}\right) = \frac{x_{i'}}{x_i}\frac{y_j}{y_j} = \frac{x_{i'}}{x_i}$, and any $\frac{y_{j'}}{y_j}$ satisfies $\psi_{ij}\left(\frac{z_{ij'}}{z_{ij}}\right) = \frac{x_i}{x_i}\frac{y_{j'}}{y_j} = \frac{y_{j'}}{y_j}$. So, it maps onto all the generators of $\mathbb{Z}\left[\frac{x_0}{x_i}, ..., \frac{x_m}{x_i}, \frac{y_0}{y_j}, ..., \frac{y_n}{y_j}\right]$, showing ψ_{ij} is surjective.

Which, this defines a closed immersion $\psi^*_{ij} : \operatorname{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, ..., \frac{x_m}{x_i}, \frac{y_0}{y_j}, ..., \frac{y_n}{y_j}\right]\right) = D_+(x_i) \times_{\operatorname{Spec}(\mathbb{Z})} D_+(y_j) \hookrightarrow \operatorname{Spec}\left(\mathbb{Z}\left[\frac{z_0}{z_{ij}}, ..., \frac{z_N}{z_{ij}}\right]\right) = D_+(z_{ij})$ (as the ring homomorphism inducing it is surjective). Upon gluing the morphisms using the structure of $\mathbb{P}^N_{\mathbb{Z}}$, this guarantees a closed immersion $\mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}} \hookrightarrow \mathbb{P}^N_{\mathbb{Z}}$ (where $N = nm+n+m$).

With the existence of Segre Embedding for projective spaces of $\mathbb{Z}$, say $s : \mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}} \hookrightarrow \mathbb{P}^{nm+n+m}_{\mathbb{Z}}$ (a closed immersion specifically), then with $N \coloneqq nm+n+m$ consider the following commutative diagram:

$$\begin{tikzcd}
\mathbb{P}^m_V \times_V \mathbb{P}^n_V \arrow[r, "p"] \arrow[d, dotted, "\exists! s'"] \arrow[dd, bend right=60, "\pi_{V,n} \circ (\pi_{V,m})'"'] & \mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}} \arrow[d, hook, "s"] \\
\mathbb{P}^N_V \arrow[r] \arrow[d, "\pi_{V,N}"] & \mathbb{P}^N_{\mathbb{Z}} \arrow[d] \\
V \arrow[r] & \operatorname{Spec}(\mathbb{Z})
\end{tikzcd}$$

Then, the universality of $\mathbb{P}^N_V = V \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^N_{\mathbb{Z}}$ guarantees a unique morphism $s' : \mathbb{P}^m_V \times_V \mathbb{P}^n_V \to \mathbb{P}^N_V$. Finally, we claim that s' is a closed immersion:

Notice in the above diagram, the tall rectangle involving $\mathbb{P}^m_V \times_V \mathbb{P}^n_V, \mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}, V, \operatorname{Spec}(\mathbb{Z})$ forms a fibre square by suitable diagram chase.

Then, this in fact enforces $\mathbb{P}^m_V \times_V \mathbb{P}^n_V \cong \mathbb{P}^N_V \times_{\mathbb{P}^N_{\mathbb{Z}}} \left(\mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}\right)$: Given any scheme W together with two morphisms $w_1 : W \to \mathbb{P}^N_V$ and $w_2 : W \to \mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}$, such that the following holds:

$$\begin{tikzcd}
W \arrow[dr, dotted, "\exists! w"] \arrow[drr, bend left, "w_2"] \arrow[ddr, bend right, "w_1"'] & & \\
& \mathbb{P}^m_V \times_V \mathbb{P}^n_V \arrow[r, "p"] \arrow[d, dotted, "\exists! s'"] & \mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}} \arrow[d, hook, "s"] \\
& \mathbb{P}^N_V \arrow[r] \arrow[d, "\pi_{V,N}"] & \mathbb{P}^N_{\mathbb{Z}} \arrow[d] \\
& V \arrow[r] & \operatorname{Spec}(\mathbb{Z})
\end{tikzcd}$$

Which, the existence and uniqueness of $w : W \to \mathbb{P}^m_V \times_V \mathbb{P}^n_V$ is guaranteed by the tall fibre rectangle, showing the top square is a fibre square. Since this characterizes $\mathbb{P}^m_V \times_V \mathbb{P}^n_V$ as the fibre product $\mathbb{P}^N_V \times_{\mathbb{P}^N_{\mathbb{Z}}} \left(\mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}}\right)$, then $s' : \mathbb{P}^m_V \times_V \mathbb{P}^n_V \to \mathbb{P}^N_V$ is a base change of $s : \mathbb{P}^m_{\mathbb{Z}} \times_{\operatorname{Spec}(\mathbb{Z})} \mathbb{P}^n_{\mathbb{Z}} \hookrightarrow \mathbb{P}^N_{\mathbb{Z}}$ (by the morphism $\mathbb{P}^N_V \to \mathbb{P}^N_{\mathbb{Z}}$), showing s' is a closed immersion (since s is).

Finally, we get this massive commutative diagram:

$$
\begin{tikzcd}
T \arrow[r, hook, "\iota_T"] \arrow[d, "\rho"'] & \mathbb{P}^n_U \arrow[r, hook, "(\iota_U)'"] \arrow[dl, "\pi_{U,n}"] & \mathbb{P}^n_{\mathbb{P}^m_V} \cong \mathbb{P}^m_V \times_V \mathbb{P}^n_V \arrow[r, hook, "s'"] \arrow[dl, "(\pi_{V,n})'"] & \mathbb{P}^N_V \arrow[ddlll, bend left, "\pi_{V,N}"] \\
U \arrow[r, hook, "\iota_U"] \arrow[d, "\varphi"'] & \mathbb{P}^m_V \arrow[dl, "\pi_{V,m}"] & & \\
V & & &
\end{tikzcd}
$$

Which, $\iota_T, (\iota_U)', s'$ are all constructed as closed immersions, hence the composition $T \hookrightarrow \mathbb{P}^N_V$ is also a closed immersion. This realizes $\varphi \circ \rho : T \to V$ as a projective morphism, hence composition of projective morphisms are still projective.

Finally, back to the original problem, given $f : X \to S$ and $g : Y \to S$ two projective morphisms, (a) guarantees their pullbacks to be projective morphisms also. So, $f' : X \times_S Y \to Y$ and $g' : X \times_S Y \to X$ are projective morphisms, satisfying the following fibre square:

$$
\begin{tikzcd}
X \times_S Y \arrow[r, "f'"] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\
X \arrow[r, "f"'] & S
\end{tikzcd}
$$

Then, (b) guarantees the composition of projective morphisms to be projective, hence $f \circ g' = g \circ f' : X \times_S Y \to S$ is projective, finishing our statement.

3 D

Problem 3. Let $f : X \to Y$ be a morphism of separated scheme of finite type over a Noetherian scheme S. Let $Z \subset X$ be a closed subscheme which is proper over S. Show that $f(Z)$ is closed in Y.

Solution. To show $f(Z) \subseteq Y$ is closed, we'll show that with the closed immersion $\iota : Z \hookrightarrow X$, one has $f \circ \iota : Z \to Y$ is proper, which will show $f \circ \iota(Z) = f(Z)$ is closed in Y.

First, let $\varphi_X : X \to S$ and $\varphi_Y : Y \to S$ be the morphism that define X, Y as separated schemes of finite type over S, this implies both φ_X, φ_Y are of finite type and separated morphisms.

Also, consider the closed immersion $\iota : Z \hookrightarrow X$, since Z is proper over S, we have this morphism $\varphi_X \circ \iota : Z \to S$ be proper.

Now, since $f : X \to Y$ is a morphism of schemes over S, we have the following commutative diagram:

$$\begin{tikzcd}
Z \arrow[dr, hook, "\iota"'] & & & \\
& X \arrow[rr, "f"] \arrow[dr, "\varphi_X"'] & & Y \arrow[dl, "\varphi_Y"] \\
& & S &
\end{tikzcd}$$

Or, $\varphi_X = \varphi_Y \circ f$, which implies $\varphi_X \circ \iota = \varphi_Y \circ (f \circ \iota)$. Since $\varphi_X \circ \iota$ is proper, with φ_Y being separated, we have $f \circ \iota$ being proper also (which is a closed map). With Z being closed in itself, $f \circ \iota(Z) = f(Z) \subseteq Y$ is closed. This finishes our proof.

4 D

Problem 4. Let k be a field and let $R = k[X,Y,Z]/I$, where $I = (X - YZ, XZ - Y^2)$. Let $W = \operatorname{Spec}(R)$.

(a) Is W a reduced scheme? Justify your answer.

(b) Is W irreducible? If not, what are its irreducible components?

(c) Prove or disprove that W has infinitely many rational points if k is infinite (recall that a point $x \in W$ is called rational point if the canonical map $k \to k(x)$ is an isomorphism, where $k(x)$ is the residue field of the local ring $\mathcal{O}_{W,x}$).

Solution. For the sake of working through all subproblems, we'll first do some calculations regarding two ideals of $k[X,Y,Z]$, defined by $P = (X,Y)$ and $Q = (X - Z^3, Y - Z^2)$. In particular, we claim that they're prime ideals that contains I:

- For $P = (X,Y)$, it's clear that $k[X,Y,Z]/\mathbb{P} \cong k[Z]$, hence it's prime; one the other hand, one has $X, Y \in P$ implies $YZ, XZ, Y^2 \in P$, hence $X - YZ, XZ - Y^2 \in P$, showing $I \subseteq P$.

- For Q, we'll show that $k[X,Y,Z]/Q \cong k[T]$ also: Consider the ring homomorphism $\varphi : k[X,Y,Z] \to k[T]$ by $\varphi(f(X,Y,Z)) = f(T^3, T^2, T)$. Which, this morphism is surjective, as any polynomial $f(T) \in k[T]$ also appears in $k[X,Y,Z]$ (say denotes as $f(Z)$), then $\varphi(f(Z)) = f(T)$. This shows that $k[X,Y,Z]/\ker(\varphi) \cong k[T]$.

  Now, we show that $Q = \ker(\varphi)$: For the generators of Q, we have $\varphi(X - Z^3) = T^3 - (T)^3 = 0$, and $\varphi(Y - Z^2) = T^2 - (T)^2 = 0$, hence $Q \subseteq \ker(\varphi)$ (as its generators are contained in ther kernel).

  For the converse, we claim that all polynomials can be expressed in the form $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ for some $f, g \in k[X,Y,Z]$, and $h(Z) \in k[Z]$: Given any monomial $X^n Y^m Z^p$, we can rewrite it as the following:

$$\begin{aligned} X^n Y^m Z^p &= \left((X - Z^3) + Z^3\right)^n \left((Y - Z^2) + Z^2\right)^m Z^p \\ &= \left(\sum_{k=0}^{n} \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k}\right) \left(\sum_{l=0}^{m} \binom{m}{l} (Y - Z^2)^l (Z^2)^{m-l}\right) Z^p \\ &= \left(Z^{3n} + \sum_{k=1}^{n} \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k}\right) \left(Z^{2m} + \sum_{l=1}^{m} (Y - Z^2)^l (Z^2)^{m-l}\right) Z^p \end{aligned} \tag{4.1}$$

  Which, define $f := \sum_{k=1}^{n} \binom{n}{k} (X - Z^3)^{k-1} (Z^3)^{n-k}$, and $g := \sum_{l=1}^{m} (Y - Z^2)^{l-1} (Z^2)^{m-l}$, the above can be rewritten as follow:

$$\begin{aligned} X^n Y^m Z^p &= (Z^{3n} + f \cdot (X - Z^3))(Z^{2m} + g \cdot (Y - Z^2)) Z^p \\ &= (f(Z^{2m} + g(Y - Z^2)) Z^p) \cdot (X - Z^3) + (g Z^{3n+p}) \cdot (Y - Z^2) + Z^{3n+2m+p} \end{aligned} \tag{4.2}$$

  This shows that each monomial can be written as some $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + Z^q$. Hence, with all polynomials being finite k-linaer combinations of the monomials, they can all be written in the form of $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ for $f, g \in k[X,Y,Z]$ and $h(Z) \in k[Z]$.

  As a consequence, for any polynomial $\ell(X,Y,Z) \in \ker(\varphi)$, since there exists $f, g \in k[X,Y,Z]$ and $h(Z) \in k[Z]$, such that $\ell(X,Y,Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$, then the evaluation by φ gives the following:

$$\varphi(\ell(X,Y,Z)) = \varphi(f \cdot (X - Z^3)) + \varphi(g \cdot (Y - Z^2)) + \varphi(h(Z)) = h(T) = 0 \tag{4.3}$$

(Note: recall that $X - Z^3, Y - Z^2 \in \ker(\varphi)$ are verified). Hence, one must have $h(T) = 0$, or $h(Z) = 0$, showing that $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) \in Q$. This proves that $\ker(\varphi) \subseteq Q$, finishing the proof of $\ker(\varphi) = Q$.

Then, since $k[X, Y, Z]/Q \cong k[T]$ based on the homomorphism φ, then Q is a prime ideal.
Finally, to say $I \subseteq Q$, use the algorithm proposed above, some calculation shows the following:

- $$X - YZ = X - ((Y - Z^2) + Z^2)Z = (X - Z^3) + Z(Y - Z^2) \in Q \tag{4.4}$$
- $$\begin{aligned} XZ - Y^2 &= XZ - ((Y - Z^2) + Z^2)^2 \\ &= XZ - \left((Y - Z^2)^2 + 2Z^2(Y - Z^2) + Z^4\right) \\ &= XZ - Z^4 - ((Y - Z^2) + 2Z^2)(Y - Z^2) \\ &= Z(X - Z^3) + (Y + Z^2)(Y - Z^2) \in Q \end{aligned} \tag{4.5}$$

Hence, the generators of I are in Q, showing $I \subseteq Q$.

Now, since $W = \operatorname{Spec}(R) \cong V(I)$ as closed subschemes of $\operatorname{Spec}(k[X, Y, Z])$, we'll work in this regime.
Here are the solutions to the subproblems:

(a) Here, we claim that W is reduced, by proving that R is reduced. In particular, we'll show that $I = P \cap Q$ to show that it's a radical.

It's clear that $I \subseteq P \cap Q$ based on the previous calculation. To show that $P \cap Q \subseteq I$, we'll consider the algorithm used when proving $k[X, Y, Z]/Q \cong k[T]$.

First, for any polynomial $s \in P \cap Q$, since $s \in P = (X, Y)$, so it can be written in the form $s = v_1 X + v_2 Y$ for some $v_1, v_2 \in k[X, Y, Z]$.

Then, perform the algorithm for Q on v_1, v_2, there exists $f_1, f_2, g_1, g_2 \in k[X, Y, Z]$, and $h_1(Z), h_2(Z) \in k[Z]$, such that the following holds:

$$\begin{aligned} s &= v_1 X + v_2 Y \\ \mathring{a} &= (f_1(X - Z^3) + g_1(Y - Z^2) + h_1(Z))X + (f_2(X - Z^3) + g_2(Y - Z^2) + h_2(Z))Y \\ &= f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y \end{aligned} \tag{4.6}$$

Which, the first four terms in the sum is in PQ (since they're multiples of products of the generators of $P = (X, Y), Q = (X - Z^3, Y - Z^2)$). So, in case for $s \in P \cap Q$, one needs $h_1(Z)X + h_2(Z)Y \in P \cap Q$.

It's true that $h_1(Z)X + h_2(Z)Y \in P$, hence the only condition is $h_1(Z)X + h_2(Z)Y \in Q$. Which, we can rewrie it as follow:

$$\begin{aligned} h_1(Z)X + h_2(Z)Y &= h_1(Z)((X - Z^3) + Z^3) + h_2(Z)((Y - Z^2) + Z^2) \\ &= h_1(Z)(X - Z^3) + h_2(Z)(Y - Z^2) + (h_1(Z)Z^3 + h_2(Z)Z^2) \end{aligned} \tag{4.7}$$

Which, one requires $h_1(Z)Z^3 + h_2(Z)Z^2 \in Q = (X - Z^3, Y - Z^2)$. Then, this imposes that $h_1(Z)Z^3 + h_2(Z)Z^2 = 0$ (as it's independent from X, Y). So, one has $Z^2(h_1(Z)Z + h_2(Z)) = 0$, showing $h_2(Z) = -h_1(Z)Z$.

As a result, the term $h_1(Z)X + h_2(Z)Y$ becomes the following:

$$h_1(Z)X - h_1(Z)YZ = h_1(Z)(X - YZ) \in I \tag{4.8}$$

So, this shows that the last two terms in $s = f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y$ is contained in I. To show $s \in I$, one simply needs the first four terms to be in I also.

Since $PQ = (X, Y)(X - Z^3, Y - Z^2) = ((X - Z^3)X, (Y - Z^2)X, (X - Z^3)Y, (Y - Z^2)Y)$, it suffices to prove these generators are in I:

- $$\begin{aligned}(X - Z^3)X &= X^2 - (XZ)Z^2 = X^2 - (XZ)Z^2 + -Y^2Z^2 + Y^2Z^2 \\ &= (X^2 - (YZ)^2) - Z^2(XZ - Y^2) \\ &= (X + YZ)(X - YZ) = Z^2(XZ - Y^2) \in I\end{aligned} \tag{4.9}$$

- $$\begin{aligned}(Y - Z^2)X &= XY - (XZ)Z = XY - (XZ)Z + Y^2Z - Y^2Z \\ &= Y(X - YZ) - Z(XZ - Y^2) \in I\end{aligned} \tag{4.10}$$

- $$\begin{aligned}(X - Z^3)Y &= XY - YZ^3 + XZ^2 - XZ^2 \\ &= Z^2(X - YZ) + XY - XZ^2 - Y^2Z + Y^2Z \\ &= Z^2(X - YZ) + Y(X - YZ) - Z(XZ - Y^2) \\ &= (Z^2 + y)(X - YZ) - Z(XZ - Y^2) \in I\end{aligned} \tag{4.11}$$

- $$\begin{aligned}(Y - Z^2)Y &= Y^2 - YZ^2 - XZ + XZ \\ &= Z(X - YZ) - (XZ - Y^2) \in I\end{aligned} \tag{4.12}$$

This shows the generators of PQ are in I, hence $PQ \subseteq I$. As a consequence, the first four terms of $s = f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y$ is also in I.

This proves $s \in I$, showing $P \cap Q \subseteq I$. Hence, $I = P \cap Q$, showing it's a radical. This shows that $R = k[X, Y, Z]/I$ is reduced, hence $W = \mathrm{Spec}(R)$ is also reduced.

(b) We show that W is reducible, in particular that $W = V(I) = V(P) \cup V(Q)$, where $V(P), V(Q)$ are its irreducible components.

First, since $I \subseteq P$ and $I \subseteq Q$, it's clear that $V(P), V(Q) \subseteq V(I)$, hence $V(P) \cup V(Q) \subseteq V(I)$; to claim the reverse inclusion, we'll consider the following element:

$$Z(X - YZ) - (XZ - Y^2) = XZ - YZ^2 - XZ + Y^2 = Y(Y - Z^2) \in I \tag{4.13}$$

Suppose $K \in V(I)$ is a prime ideal containing I, then $Y(Y - Z^2) \in K$, which the prime property guarantees $Y \in K$ or $(Y - Z^2) \in K$.

- If $Y \in K$, then since $X - YZ \in I \subseteq K$, one has $X = (X - YZ) + YZ \in K$, showing $P = (X, Y) \subseteq K$, or $K \in V(P)$.
- Else if $Y - Z^2 \in K$, then since $X - YZ \in I \subseteq K$ again, one has $X - Z^3 = (X - YZ) + (YZ - Z^3) = (X - YZ) + Z(Y - Z^2) \in K$, showing $Q = (X - Z^3, Y - Z^2) \subseteq K$, or $K \in V(Q)$.

Hence, in either case $K \in V(P) \cup V(Q)$, showing $W = V(I) = V(P) \cup V(Q)$.

To show that W is reducible, we'll show that $V(P), V(Q)$ are proper closed subsets of W: Suppose the contrary that one of them is not proper closed subset of W, there are two cases:

- If $V(P) = W$, this shows that $V(Q) \subseteq V(P)$; with P, Q being prime (which both are radicals), this shows that $Q \supseteq P$. However, if consider the ring homomorphism $\psi : k[X, Y, Z] \twoheadrightarrow k$ by $\psi(X, Y, Z) = \psi(1, 1, 1)$, notice that $\psi(X - Z^3) = 1 - 1^3 = 0$ and $\psi(Y - Z^2) = 1 - 1^2 = 0$, showing $Q = (X - Z^3, Y - Z^2) \subseteq \ker(\psi)$; yet, if plugin the generators of P, we get $\psi(X) = \psi(Y) = 1 \neq 0$, showing $P \nsubseteq \ker(\psi)$, which contradicts our construction that $\ker(\varphi) \supseteq Q \supseteq P$.

- Else if $V(Q) = W$, this shows that $V(P) \subseteq V(Q)$, which implies $P \supseteq Q$. Howeve, consider the ring homomorphism $\rho : k[X,Y,Z] \twoheadrightarrow k$ by $\rho(f(X,Y,Z)) = f(0,0,1)$, notice that $\rho(X) = \rho(Y) = 0$, showing $P = (X,Y) \subseteq \ker(\rho)$; yet, if plugin the generators of Q, we get $\rho(X - Z^3) = 0 - 1^3 = -1 \neq 0$, showing $Q \not\subseteq \ker(\rho)$, again this contradicts our construction that $\ker(\rho) \supseteq P \supseteq Q$.

Since in either case we get a contradiction, then $V(P), V(Q)$ cannot be the whole W, showing they're proper closed subsets, which proves that $W = V(P) \cup V(Q)$ is reducible.

Finally, to show $V(P), V(Q)$ are its irreducible components, consider any other irreducible subsets $W' \subseteq W = V(I)$, since W' is also closed in $\mathrm{Spec}(k[X,Y,Z])$ (by the closeness of $V(I)$ in it), then $W' = V(K)$ for some ideal $K \subseteq k[X,Y,Z]$. Which, the irreducibility of W' guarantees K to be a prime ideal. However, if $V(K) \subseteq V(I)$, one has $I \subseteq \sqrt{I} \subseteq K$ (by primeness, it is a radical), showing that $K \in V(I) = V(P) \cup V(Q)$. This implies that either $K \in V(P)$ (implying $P \subseteq K$, or $V(P) \supseteq V(K)$), or $K \in V(Q)$ (implying $Q \subseteq K$, or $V(Q) \supseteq V(K)$), showing that $W' = V(K)$ is contained in $V(P)$ or $V(Q)$. Hence, these two must be the irreducible components of W.

(c) With k being infinite, we'll show that there are infinitely rational points in W. For any $t \in k$, consider the maximal ideal $\mathfrak{m}_t := (X, Y, Z - t) \subset k[X,Y,Z]$. Notice that it contains I, since $X - YZ, XZ - Y^2 \in \mathfrak{m}_t$ by the fact that $X, Y \in \mathfrak{m}_t$. Hence, $x := \mathfrak{m}_t$ can be realized as a maximal ideal in $R = k[X,Y,Z]/I$ by descending to the quotient, which is realized as a closed point in $W = \mathrm{Spec}(R)$.

To compute its residue field, let $\overline{\mathfrak{m}}_t \subseteq R$ denotes the quotient of $\mathfrak{m}_t$. Then, its corresponding residue field is as follow:

$$k(\overline{\mathfrak{m}}_t) = \frac{R_{\overline{\mathfrak{m}}_t}}{\overline{\mathfrak{m}}_t R_{\overline{\mathfrak{m}}_t}} \cong \left(\frac{R}{\overline{\mathfrak{m}}_t}\right)_{\overline{\mathfrak{m}}_t} \cong k \tag{4.14}$$

Note that since $I \subseteq \mathfrak{m}_t$, then the composition of projections $\varphi : k[X,Y,Z] \twoheadrightarrow R \twoheadrightarrow R/\overline{\mathfrak{m}}_t$ has the kernel being precisely $\varphi^{-1}(\overline{\mathfrak{m}}_t) = \mathfrak{m}_t$, hence $k \cong k[X,Y,Z]/\mathfrak{m}_t \cong R/\overline{\mathfrak{m}}_t$ (where, the first isomorphism can be realized by the evaluation $e : k[X,Y,Z] \twoheadrightarrow k$ by $e(f(X,Y,Z)) = f(0,0,t)$, where the kernel is precisely $\mathfrak{m}_t = (X, Y, Z - t)$).

Hence, with k being infinite, there are infinitely many $t \in k$, with $\overline{\mathfrak{m}}_t \subseteq R$ being a maximal ideal (or a closed point in $W = \mathrm{Spec}(R)$), such that the residue field is isomorphic to k. Hence, W has infinitely many rational points.

5 D

Problem 5. Let $f : X \to Y$ be a surjective morphism between two schemes which are finite type and separated integral schemes over a field. Assume that X is an affine scheme. Prove or disprove that Y is an affine scheme if and only if it is a quasi-affine scheme (recall that quasi-affine means open subscheme of an affine scheme).

Solution. We'll prove that Y is affine $\iff$ Y is quasi-affine.

$\Longrightarrow$:

Suppose Y is affine, then it's an open subscheme of Y (an affine scheme), hence Y is quasi-affine.

$\Longleftarrow$:

Suppose that Y is quasi-affine. Here, we'll denote the global section of Y as $A := \mathcal{O}_Y(Y)$, and let B be a commutative ring satisfies $X \cong \operatorname{Spec}(B)$ (which, X, Y being integral schemes implies A, B are integral domains).

Let's first deal with a special case: Suppose B is a field, then $X = \operatorname{Spec}(B)$ is a one point scheme, hence with $f : X \twoheadrightarrow Y$ being surjective, Y must be a one point scheme also. Take any affine neighborhood of the only point of Y, it must be Y itself, showing that $Y = \operatorname{Spec}(A')$ for some ring A' (in particular, $A' = \mathcal{O}_Y(Y) = A$). So, Y is affine.

From now one, we'll assume B (the global section of X) is not a field.

Let k be a field where X, Y are finite type and separated integral schemes over it. Which, given $\varphi_X : X \to \operatorname{Spec}(k)$, $\varphi_Y : Y \to \operatorname{Spec}(k)$ as the corresponding morphisms, φ_X, φ_Y are finite type and separated morphisms that satisfy the following diagram:

$$\begin{tikzcd}
X = \operatorname{Spec}(B) \arrow[rr, two heads, "f"] \arrow[dr, "\varphi_X"'] & & Y \arrow[dl, "\varphi_Y"] \\
& \operatorname{Spec}(k) &
\end{tikzcd}$$

Then, on the global section level, this reverses to the following diagram of ring homomorphisms:

$$\begin{tikzcd}
B & & \mathcal{O}_Y(Y) = A \arrow[ll, "f^{\#}"'] \\
& k \arrow[ul, hook, "\varphi_X^{\#}"] \arrow[ur, hook, "\varphi_Y^{\#}"'] &
\end{tikzcd}$$

Now, we claim the following statements in order:

(a) Y can be realized as open subschemes of $\operatorname{Spec}(A)$.

(b) Y is affine, in particular $Y \cong \operatorname{Spec}(A)$.

Proof of (a):

Since Y is assumed to be quasi-affine, there exists an open immersion $\iota_Y : Y \hookrightarrow \operatorname{Spec}(R)$ for some commutative ring R. Then, since the affinization of Y, say $\operatorname{Aff}_Y = \operatorname{Spec}(A)$, with the natural morphism of schemes $i_Y : Y \to \operatorname{Aff}_Y$, there exists a unique morphism of schemes $\psi_Y : \operatorname{Aff}_Y \to \operatorname{Spec}(R)$, such that the following commutative diagram holds:

$$\begin{tikzcd}
Y \arrow[rr, "i_Y"] \arrow[dr, hook, "\iota_Y"'] & & \mathrm{Aff}_Y \arrow[dl, dotted, "\exists! \psi_Y"] \\
& \mathrm{Spec}(R) &
\end{tikzcd}$$

Which, since ι_Y is injective (as set map), this enforces i_Y to also be injective also.

Now, consider the ring homomorphism $\psi_Y^\# : R \to A$. For any $P \in Y \hookrightarrow \mathrm{Spec}(R)$, there exists some element $f \in R$, such that $P \in D(f) \subseteq Y$ (since Y is open subscheme of $\mathrm{Spec}(R)$). Then, denote $f' := \varphi_Y^\#(f) \in A$. Using the Hartshorne Notation, we can consider the subset $Y_{f'} \subseteq Y$ (denoting all points $y \in Y$, such that the localization $f'_y \in \mathcal{O}_{Y,y}$ is invertible, or not in the maximal ideal $\mathfrak{m}_y \subseteq \mathcal{O}_{Y,y}$). Notice that this set is precisely $Y_{f'} = D(f)$, since when passing to the stalk, any $P \in D(f)$ has $\mathcal{O}_{Y,P}$ has f being an invertible element, hence with f' being the image of f in A, under localization it's also invertible.

As a result, one has $R_f \cong \mathcal{O}_{\mathrm{Spec}(R)}(D(f)) = \mathcal{O}_Y(D(f)) = \mathcal{O}_Y(Y_{f'}) = \mathcal{O}_Y(D(f')) \cong A_{f'}$, showing that $P \in D(f) \subseteq Y$ has an affine neighorhood $D(f) \cong D(f') \cong \mathrm{Spec}(A_{f'})$ in Y.

This shows that $i_Y : Y \hookrightarrow \mathrm{Spec}(A)$ is actually an open immersion (as any point in Y has an open neighborhood in Y, such that it's scheme-isomorphic to a fundamental open subset of $\mathrm{Spec}(A)$). So, Y is an open subscheme of $\mathrm{Spec}(A)$.

Proof of (b):

Now, together with the open immersion $i_Y : Y \hookrightarrow \mathrm{Spec}(A)$, the morphism $\varphi_Y : Y \to \mathrm{Spec}(k)$ uniquely factors through some morphism $\rho : \mathrm{Spec}(A) \to \mathrm{Spec}(k)$ (since $\mathrm{Spec}(A) = \mathrm{Aff}_Y$ the affinization), which we have the following commutative diagram of morphisms:

$$\begin{tikzcd}
X \arrow[r, two heads, "f"] \arrow[dr, "\varphi_X"'] & Y \arrow[r, hook, "i_Y"] \arrow[d, "\varphi_Y"] & \mathrm{Spec}(A) \arrow[dl, dotted, "\exists! \rho"] \\
& \mathrm{Spec}(k) &
\end{tikzcd}$$

Where the morphism $i_Y \circ f$ is separated (as it's morphism between affine schemes). Which, on the level of global section, we have the following commutative diagram:

$$\begin{tikzcd}
B & A \arrow[l, "f^\#"'] & A \arrow[l, hook, two heads, "\mathrm{id}_A"'] \\
& k \arrow[ul, hook, "\varphi_X^\#"] \arrow[u, hook, "\varphi_Y^\#"] \arrow[ur, hook, "\rho^\#"'] &
\end{tikzcd}$$

Where the middle A is given by the global section of Y, $A = \mathcal{O}_Y(Y)$.

Then, this implies that the morphism $i_Y \circ f : X = \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is induced by the ring homomorphism $f^\# \circ \mathrm{id}_A = f^\# : A \to B$; and, for the diagram to commute, $f^\#$ is in fact a k-algebra homomorphism. Moreover, $f^\#$ is of finite type, as B is a finitely generated k-algebra, and the morphism is a k-algebra homomorphism. This realizes B as a finitely generated A-algebra also.

Now, notice that $i_Y \circ f$ is a dominant morphism: Because A is an integral domain, then $\mathrm{Spec}(A)$ is irreducible, hence $Y \subseteq \mathrm{Spec}(A)$ as a open subscheme is dense and irreducible (or $\overline{Y} = \mathrm{Spec}(A)$). With f being surjective, one has $f(X) = Y$, hence $\overline{i_Y \circ f(X)} = \overline{i_Y(Y)} = \overline{Y} = \mathrm{Spec}(A)$. Hence, with the morphism $i_Y \circ f$ being dominant, the induced ring homomorphism $f^\#$ is injective.

Finally, we claim that $Y = \mathrm{Spec}(A)$: Since $Y \subseteq \mathrm{Spec}(A)$ is open, there exists $a_i \in A$ (where $i \in I$ for some index set), such that $Y = \bigcup_{i \in I} D(a_i)$. Since $\varphi_Y : Y \to \mathrm{Spec}(k)$ is of finite type (which is quasi-compact), with $\mathrm{Spec}(k)$ being quasi-compact, this implies $Y = \varphi_Y^{-1}(\mathrm{Spec}(k))$ is quasi-compact. Hence, one can choose the cover to be finite, say $a_1, ..., a_n \in A$ satisfies $Y = \bigcup_{i=1}^n D(a_i)$.

Then, with $f : X \twoheadrightarrow Y$ and the induced ring homomorphism $f^\# : A \hookrightarrow B$, one has $\mathrm{Spec}(B) = X = f^{-1}(Y) = \bigcup_{i=1}^n f^{-1}(D(a_i)) = \bigcup_{i=1}^n D(f^\#(a_i))$. So, since $D(f^\#(a_i))$ forms an open cover of $\mathrm{Spec}(B)$, this implies $B = (f^\#(a_1), ..., f^\#(a_n))$.

At this point I was aiming for $A = (a_1, ..., a_n)$ (currently in progress), then with $\bigcup_{i=1}^n D(a_i) = \operatorname{Spec}(A)$ based on this argument, it'll finish the proof that $Y = \operatorname{Spec}(A)$, or showing Y is affine.